

Essential Mathematics and Statistics for Scientists

02-680 – Fall 2026

Revised: August 26, 2026
(change since start of classes marked in orange)

Class Location and Times:

[Lecture] Tuesday & Thursday 9:30am-10:50am,
Posner Hall (POS) 152

[Recitation] Friday 5:00pm-6:20pm,
Margret Morrison (MM) 103

Course Description: This course rigorously introduces fundamental topics in mathematics and statistics to first-year master's students as preparation for more advanced computational coursework. Topics are sampled from discrete mathematics, linear algebra, calculus, and probability and statistics, with an emphasis on the tools most relevant to computational biology and machine learning. This background will also serve students well in advanced courses that apply concepts in machine learning to scientific datasets, such as 02-710 (Computational Genomics) or 02-750 (Automation of Biological Research).

Prerequisite: High school mathematics (including single- and multi-variable calculus) and ability to reason mathematically.

Course Objectives: At the end of the semester, students will be able to:

1. Reason precisely using formal mathematical language as a foundation for describing computational and biological problems.
2. Perform and interpret core linear algebra operations as they apply to real datasets.
3. Use calculus, particularly derivatives and gradients, to analyze and optimize functions.
4. Apply the axioms of probability and reason about random variables, distributions, and multivariate/graphical models.
5. Estimate model parameters from data using maximum likelihood and Bayesian methods, and quantify uncertainty in those estimates.
6. Design and interpret statistical analyses, including hypothesis tests and regression models, and communicate their conclusions and limitations.

1 Instructional Staff

Instructor

Dr. Dan DeBlasio email: deblasio@cmu.edu
 office: GHC 7707
 office hours: TBD
 appointments: calendly.deblasiolab.org

Teaching Assistants

Seo-Yoon Moon email: smoon2@andrew.cmu.edu
 office hours: Monday 1pm

Emmanuel Mhrous email: emhrous@andrew.cmu.edu
 office hours: Tuesdays, 1pm

Sorro Sun email: sorros@andrew.cmu.edu
 office hours: Tuesday, 5pm

Dylan Setiawan email: dgsetiaw@andrew.cmu.edu
 office hours: Wednesday, 11am

All TA Office Hours in GHC 7404.

2 Logistics

Course Topic Schedule: Can be found [on Google Sheets¹](#) and is subject to change.

Textbook: There are no required textbooks for this course. Below are textbooks that can be good references:

- **Mathematics and Statistics for the Quantitative Sciences** by Betti
- **Connecting Discrete Mathematics and Computer Science** by Liben-Nowell²
- **Mathematics for Machine Learning** by Deisenroth, Faisal, Ong
- **All of Statistics: A Concise Course in Statistical Inference** by Larry Wasserman
- **Probability and Statistics** by Morris DeGroot and Mark Schervish
- **Basic Concepts of Linear Algebra** by Isaak and Manougian

¹<https://docs.google.com/spreadsheets/d/1AuxF0rYH08UFzA6YwOEVaYlRdU7XbZr-j7E7Gvi0G9s>

²<https://www.cs.carleton.edu/faculty/dln/book/>

Communication platforms:

- **Homepage.** Canvas will be used for quizzes and as a central repository for grades. You should be automatically enrolled at <https://canvas.cmu.edu/courses/54762>.
- **Discussion Forum.** An online forum is provided on Piazza as an area for discussion and questions linked within Canvas. The forum will be moderated by the course staff who will respond to questions, but students are encouraged to help each other via discussion, within the bounds of the collaboration policy (Section 5.2); hints on assignment specifics will be provided by the teaching staff rather than discussed on the forum. <https://piazza.com/cmu/fall2026/02680>.
- **Gradescope.** Linked from within Canvas, allows for submission, grading, and feedback for homework assignments. <https://www.gradescope.com/courses/1341903>.
- **NotebookLM.** Custom chat agent trained on our class notes. <https://notebook.google.com/notebook/e19fffd8-528b-4ae8-b0c2-9cda8982883c>

3 Grading

Grades are communicated to students in a timely manner. It is the students' responsibility to keep track of their grades by compiling the grades they receive. Your semester grade will be based on a combination of homework assignments, weekly quizzes, class participation, mid-term assessment, and a final exam.

The approximate percentages are as follows:

10%	Homework
10%	Attendance & Participation
30%	Quizzes
50%	Exams (25% per)

The base percentage-score-to-letter-grade conversion for this course is as follows:

	+		-
A	98%	93%	90%
B	88%	83%	80%
C	78%	73%	70%
D	68%	63%	—

Any lower scores will result in a grade of “R”. These minimums may be lowered without notice but will not be raised.

3.1 Homework

Written homework assignments will test your knowledge of the material covered in class. Homework assignment will be made available at least one week before the due date.

All writing assignments are to be submitted on Canvas (via Gradescope) on the day it is due. Any assignment turned in after the due date will be deducted 5 percentage points for the first day it is late, and 20 percentage points for every day after that. If you have extenuating circumstances, just communicate with us and we'll do what we can to accommodate.

Homework will be submitted as a pdf document which will need to be generated using an editor of the student's choosing. Due to the reliance of this course on mathematical notation the use of L^AT_EX is encouraged. If using another editor, students should utilize the editor's equation editor to ensure the correct answers are being communicated. To assist those students using L^AT_EX, the source code for each assignment will be provided in addition to the pdf.

Cheating policy

All work must be your own. Unauthorized collaboration or plagiarism will result in a negative grade (e.g., a homework worth 100 points will be factored in as a -100 points towards your final grade) and will be reported to your academic advisor and dean. See Section 5.

3.1.1 Regrading of Assignments

In the event that you feel an assignment was graded incorrectly, you must make the request in writing within one week of receiving the graded material. Your request should include a reference to the section of the assignment you believe was incorrectly graded. By asking for your assignment to be regraded, the whole assignment will be regraded and it is possible that fewer points may be awarded than the previous grade in light of new consideration.

3.2 Attendance & Participation

Attendance will *not* be taken, but we will have occasional in-class exercises that serve to reinforce the concepts we have covered and provide feedback to the instructor. These exercises will not be graded for correctness, but participation will be expected in order to receive credit. You are allowed three “dropped” attendance grades without penalty. These can be used for any purpose.

All classes will be live without an option for virtual participation.

In class activities will be weighted as follows: assuming a is the fraction of activities participated in, the grade will be calculated as

$$\text{participation} = 10\% \cdot \frac{a}{0.75}$$

meaning students only need to participate in 75% of activities to receive full credit, and

can receive up to 3% extra credit for participating in more.

3.3 Quizzes

There will be a 10 minute quiz at the start of each Tuesday session. This will cover topics from the previous week. For those with the prerequisite knowledge, the quiz can be taken before the weeks content by scheduling a time at [calendly.dandebblasio.com/02-680-quizzes](https://calendly.com/dandebblasio/02-680-quizzes). The quiz must be taken before the content in class, and will count to cover both in-class activity credit as well as the quiz.

As a timing example: week 3 (Sept 8 and 10) is planned to cover Graphs and Trees, a quiz on this content will be at the start of class Tuesday of week 4 (Sept 15). If you feel you know all of the content to be presented, you can take the quiz during week 2 (Aug 31-Sept 3) and if passed you will also get participation credit.

	Monday	Tuesday	Wednesday	Thursday	Friday
week -1	pre-quiz slots	pre-quiz slots	pre-quiz slots	pre-quiz slots	
content week		content		content	recitation
following week		in-class quiz			

Students will be allowed 3 quiz tokens through the semester (computed as dropping the lowest 3 scores). Taking the in-class quiz will overwrite any pre-quiz score (even if it is lower) and void the in-class activity waiver for the week.

3.4 Exams

The two exams will test your knowledge of the material from the class. The first midterm will be held in class, and the second midterm will be held during the university's scheduled time. Dates for the exams will be announced in class, and can be found on the course schedule when available.

The exams will not be cumulative: midterm 2 will cover material encountered after midterm 1. That having been said, later material in the class may build upon the techniques covered in the earlier material.

4 Expectations

- Please note, the use of cell phones is not permitted during lecture. These should be left in your backpack or not brought to class.
- Laptop computers may be used for taking notes or for in-class assignments but cannot be used for other activities during lecture.
- No student may record any classroom activity without the express written consent of the instructor. If you believe you need to record classroom activities as an accommodation, please contact the Office of Disability Resources (see Section ??).

Please keep in mind that these guidelines are necessary to maintain an environment that is safe and conducive for learning.

4.1 Expectations of Students and Instructors

The instructors and teaching assistants have the right to expect the following of students: (Adapted and modified from those developed by [Howard Culbertson at Southern Nazarene University](#)).

1. Students will arrive to class on time and will be prepared for the lecture.
2. Students will turn in assignments on time (see policy on assignment due dates).
3. Students will immediately inform the instructor or the teaching assistants if extenuating circumstances prevent the student from attending a lecture.
4. Students will follow the code of conduct regarding academic integrity, cheating, plagiarism, and collaboration as outlined in the syllabus.
5. Students will seek assistance when they need it.
6. If contacted by the instructor or teaching assistant, students will respond within 24 hours during the week and 48 hours on weekends.

The students have the right to expect the following of the instructor and teaching assistants:

1. A syllabus that describes class procedures, policies, and a course description will be provided.
2. Class sessions that will start and end on time.
3. Any changes to the course schedule will be provided to the students within 48 hours of the change.
4. The instructor will be available outside class either during their posted office hours or during other pre-arranged times.
5. If contacted by a student, the primary instructors or teaching assistants will respond within 24 hours during weekdays and 48 hours on weekends.

5 Academic Integrity

(See: <https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>)

Students in this course are expected to meet CMU's academic integrity standards in full — see the full university policy linked above. In short, this means submitting your own work, giving credit for others' ideas and contributions, and maintaining the trust that makes an academic community possible. Students who cannot meet these standards should voluntarily withdraw from this course.

5.1 Use of Generative AI in this course

To best support your own learning (remember this course builds the quantitative foundation the rest of your program assumes you have) you must complete all graded work in this course entirely on your own, without the use of generative artificial intelligence (AI). This is a blanket policy: it applies to every graded assignment and classroom exercise, including homework, quizzes, and exams, with no exceptions.

You may not use AI tools to generate any content (text, code, derivations, proofs, explanations, or any other material) that you then submit as your own work. Cutting and pasting AI-generated content into an assignment, or paraphrasing it before submitting, is a violation of CMU's Academic Integrity Policy, exactly as if the material had come from another student or an uncredited outside source. (This policy covers generative AI tools such as ChatGPT, Claude, Copilot, and Gemini — it does not cover ordinary calculators or non-generative computational tools.)

Using AI as a study aid is encouraged. You're welcome to use generative AI, including the course's shared NotebookLM, to help you understand concepts, work through confusion while studying, or explore the material from a different angle. The line is between using AI to learn the material and using it to produce the work you submit for a grade: the first is fine, the second is not, no matter how you use, phrase, or rework the output.

If you're ever unsure whether a particular use crosses that line, ask before you use it, not after.

Suspected violations will be handled under CMU's Academic Integrity Policy³.

5.2 Cheating and Plagiarism

In order to deter and detect plagiarism, online tools and other resources may be used in this class. It is the ethical responsibility of each student to identify the conceptual sources of any work submitted; failure to do so is dishonest and is the basis for a charge of cheating or plagiarism, which is subject to disciplinary action.

Cheating includes but is not necessarily limited to:

1. Plagiarism, explained below.
2. Submission of work that is not the student's own for papers, assignments, or exams.
3. Submission or use of falsified data.
4. Theft of or unauthorized access to an exam.
5. Use of an alternate, stand-in or proxy during an examination.
6. Use of unauthorized material including textbooks, notes or computer programs in the preparation of an assignment or during an examination.
7. Supplying or communicating in any way unauthorized information to another student for the preparation of an assignment or during an examination.

³<https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>

8. Unauthorized collaboration in the preparation of an assignment (see “Collaboration vs. Cheating,” Section 5.2, below).
9. Submission of the same work for credit in two courses without obtaining the permission of the instructors beforehand.

Plagiarism includes, but is not limited to, failure to indicate the source with quotation marks or footnotes where appropriate if any of the following are reproduced in the work submitted by a student:

1. A phrase, written or musical.
2. A graphic element.
3. A proof.
4. Specific language.
5. An idea derived from the work, published or unpublished, of another person.

Any disciplinary actions regarding charges of cheating or plagiarism will follow the procedures of the home university of the student involved.

Collaboration vs. Cheating Collaboration is defined by Merriam-Webster’s Collegiate Dictionary (10th edition) as “to work jointly with others or together, especially in an intellectual endeavor.” You are encouraged to discuss the course material, concepts, and homework assignments with other students in the class. **However, each student must eventually submit their own unique work (e.g., homework solutions, final project).** If any collaboration was used to complete an assignment, record the names of the collaborators and the nature of the collaboration. Any attempt to submit work that is not the student’s own work will be considered an act of cheating. In addition, any student who knowingly supplies their completed homework assignment to another student is violating the cheating policy and will be subject to disciplinary action.

ANY VIOLATION OF THIS POLICY WILL NOT BE TOLERATED AND THE PENALTY WILL BE FAILURE IN THE COURSE.

If you have any questions regarding this policy, contact the instructors.

6 Resources

6.1 Accommodations for Students with Disabilities

If you have an accommodations letter from the Office of Disability Resources, please discuss your accommodations and needs with me as early in the semester as possible so we can make sure they’re in place. If you suspect you may have a disability and would benefit from accommodations but are not yet registered, contact the Office of Disability Resources directly: cmu.edu/disability-resources/ (access@andrew.cmu.edu).

6.2 Statement of Support for Students' Health & Well-being

Take care of yourself. If you or anyone you know is experiencing academic stress, difficult life events, or feelings of anxiety or depression, please reach out for support sooner rather than later — to a friend, to me, or to CaPS.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

Counseling & Psychological Services (CaPS): 412-268-2922, cmu.edu/counseling/
Suicide and Crisis Lifeline: 988 Re:solve Crisis Network: 888-796-8226

If the situation is life threatening, call the police:
412-268-2323 (on campus) or 911 (off campus)

6.3 Inclusivity

We must treat every individual with respect. Diversity (race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, ancestry, belief, veteran status, genetic information, etc.) is fundamental to building and maintaining an inclusive campus community; each of us is responsible for creating a safer, more inclusive environment. See CMU's full inclusivity resources at cmu.edu/inclusive-excellence/.

If you experience or observe unfair or hostile treatment on the basis of identity, speak to the **Center for Student Diversity and Inclusion** ([csdi@andrew...](mailto:csdi@andrew.cmu.edu), (412) 268-2150).

6.4 Ethics Reporting Hotline

To report suspected unethical or illegal activity related to academic and student life, financial matters, human relations, or campus safety — confidentially or anonymously — use CMU's Ethics Reporting Hotline, operated by a third party (NAVEX): call 1-844-587-0793 or visit cmu.ethicspoint.com. **This hotline is *not* an emergency service.**